



DOSM Datathon 2026

TourismBalance AI — Dynamic Tourism Resilience & Demand Intelligence

by

The Outliers

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Contents

Contents	2
Executive Summary	4
1. Introduction.....	5
1.1 Background	5
1.2 Problem Statement	5
1.3 Objectives.....	5
1.4 Scope and Non-Scope	5
2. Literature Review.....	6
2.1 Tourism Demand and Resilience	6
2.2 Transport Access as a Tourism-Relevant Proxy	6
2.3 Time-Series Machine Learning and Baselines	6
2.4 Decision Support	6
2.5 Literature Gap and Reference Control	6
3. Conceptual Framework.....	7
4. Data and Destination Mapping	7
4.1 Data Sources.....	7
4.2 Destination Mapping	8
4.3 Final ML Targets.....	8
5. Data Preparation and Feature Engineering	10
5.1 Cleaning and Coverage	10
5.2 Frozen Feature Set.....	10
5.3 Leakage Control	10
6. Forecasting and Evaluation Methodology	10
6.1 Candidate Models.....	10
6.2 Direct Multi-Horizon Design	10
6.3 Evaluation Metrics	11

6.4 Demand Classes	12
6.5 Recommendation Logic	12
7. Results and Findings	13
7.1 Final Test Performance	13
7.2 Verified Positive Results	14
7.3 Negative and Neutral Results	14
7.4 Horizon Stability	14
7.5 Data Recency	14
8. Output (Dashboard): Power BI Decision-Support System	15
8.1 Executive Overview	15
8.2 State Tourism Intelligence	16
8.3 AI Mobility Demand Forecast	18
8.4 Smart Travel Recommendation	19
8.5 ML Performance & Model Evaluation	20
9. Feature Importance	21
10. Limitations	21
11. Future Work	21
12. Conclusion	22
References	23

Executive Summary

TourismBalance AI is an implemented decision-support prototype that connects state-level tourism intelligence with tourism-relevant rail mobility forecasting. Its analytical story is UNDERSTAND → PREDICT → RECOMMEND → EVALUATE: understand where tourism activity is concentrated, predict passenger movements for validated destination–station links, recommend lower-demand arrival windows or days, and evaluate every selected model against a weekly baseline.

The final system combines official tourism indicators, public rail-ridership records, and a manually curated destination mapping. The state intelligence layer reports Domestic Visitors, Hotel Occupancy Rate, and Visitor Intensity. The mobility layer models 15 targets: ten hourly KTMB targets across ETS, Komuter, Komuter Utara and Intercity services, plus five daily Rapid Rail targets. BRT appeared in the broader destination-mapping exercise but was excluded from the final machine-learning and dashboard scope.

Four candidates were evaluated: a naive weekly baseline, Linear Regression, Random Forest, and XGBoost. Direct Day 1–Day 7 models used chronological Train → Validation → Test partitions; validation performance selected the winner and the final test remained retrospective evaluation. Only Pasar Seni and Seremban beat the weekly baseline on the final test. The dashboard therefore reports both positive and negative results without claiming universal model superiority.

Interpretation boundary. Rail ridership represents passenger movements, not confirmed tourist counts. The forecasts are tourism-relevant mobility proxies for mapped destinations, not forecasts of tourist arrivals.

1. Introduction

1.1 Background

Tourism resilience depends on understanding both tourism activity concentration and the timing of access demand. State-level statistics show where activity is concentrated, while transport data reveals when passenger movements rise around gateways serving tourism destinations. Used together, these sources can support planning without implying that every rail passenger is a tourist.

TourismBalance AI was developed as a five-page Power BI decision-support prototype for DOSM Datathon 2026. It translates descriptive tourism indicators and experimentally forecast rail demand into an executive workflow for planners and visitors.

1.2 Problem Statement

Tourism and mobility datasets are typically published at different geographic and temporal grains. State tourism indicators are annual, whereas rail data may be daily or hourly. Destination names also do not directly match station identifiers. This fragmentation makes it difficult to connect broad tourism concentration with short-term mobility conditions, assess forecast reliability, and communicate operationally useful arrival guidance.

1.3 Objectives

Integrate authoritative tourism and rail data into a traceable analytical model.

Map selected rail stations to tourism-relevant destinations using documented evidence and confidence labels.

Forecast passenger movements for 15 validated targets over direct horizons from Day 1 to Day 7.

Classify forecast demand relative to each target's historical distribution and recommend lower-demand arrival windows or days.

Evaluate selected models against a transparent weekly baseline and communicate limitations truthfully in Power BI.

1.4 Scope and Non-Scope

The frozen model uses tourism, destination-mapping and rail-ridership inputs available in the final project files. It does not use weather, rainfall, temperature, event calendars, public holidays or school holidays. Those variables are future-work candidates only. The implementation is a prototype and does not estimate causal tourism effects.

2. Literature Review

2.1 Tourism Demand and Resilience

Official domestic-tourism statistics provide a defensible foundation for describing tourism concentration and change across Malaysian states. The implemented state layer uses Domestic Visitors and Hotel Occupancy Rate directly and derives Visitor Intensity as a population-normalised visitor-intensity proxy. These indicators answer a spatial question—where activity is concentrated—rather than a short-term forecasting question.

2.2 Transport Access as a Tourism-Relevant Proxy

Rail stations can function as access points to attractions, commercial districts, heritage areas and nature gateways. In this study, proximity and documented transport relevance support a destination–station mapping, but ridership is never relabelled as tourist volume. The distinction protects the interpretation of the dashboard.

2.3 Time-Series Machine Learning and Baselines

Time-series evaluation must respect chronology. Random train–test splits can leak future information into model development. The project therefore uses chronological partitions and evaluates direct multi-horizon forecasts. A weekly naive model is retained as a strong and interpretable benchmark because recurring weekly patterns are plausible in passenger demand.

2.4 Decision Support

Forecast values become more actionable when translated into demand classes and lower-demand arrival guidance. The dashboard also surfaces recency status, source cutoffs and baseline comparisons so that a user can judge whether an output is current enough and accurate enough for its intended use.

2.5 Literature Gap and Reference Control

Weather, holidays and events were not included in the frozen dataset or model. Their possible inclusion is identified only as future work; this study makes no claim about their effect on predictive performance.

3. Conceptual Framework

Stage	Question	Implemented output
UNDERSTAND	Where is tourism activity concentrated?	State metrics, map, ranking and trend
PREDICT	When will mapped passenger demand rise?	Direct Day 1–Day 7 forecasts for 15 targets
RECOMMEND	When is a lower-demand arrival opportunity?	Hourly window for KTMB; lower-demand day for Rapid Rail
EVALUATE	How credible is the selected model?	MAE, RMSE, WAPE and weekly-baseline comparison

The framework deliberately separates descriptive tourism intelligence from the predictive mobility layer. The tourism metric selector controls the Top State KPI, Malaysia map and state ranking only. Destination and service selectors control mobility forecasts, recommendations and performance outputs.

Core interpretation. *The system predicts tourism-relevant passenger movements at mapped rail targets. It does not predict tourist counts and does not infer that mobility demand is caused by tourism.*

4. Data and Destination Mapping

4.1 Data Sources

Source	Role	Grain / use
DOSM tourism datasets	Domestic Visitors and state context	Annual state intelligence
Tourism / accommodation context	Hotel Occupancy Rate and capacity context	Annual state intelligence
Population context	Visitor Intensity denominator	Annual visitor-intensity proxy

Source	Role	Grain / use
data.gov.my / KTMB	ETS, Komuter, Komuter Utara and Intercity ridership	Hourly ML targets
data.gov.my / Rapid Rail	Urban rail ridership	Daily ML targets
Tourism Malaysia and official destination sources	Destination relevance evidence	Mapping validation

4.2 Destination Mapping

The mapping file contains 44 destination–station records across 32 unique stations and 12 states. Each row records the rail service, station, destination, state, category, mapping reason, evidence source, URL, confidence and notes. The final distribution is 35 high-confidence and nine medium-confidence mappings. Confidence describes mapping evidence; it is not model accuracy.

Linkages were established through direct station naming, documented proximity or gateway relevance where supported by official destination information. No geographic mapping was invented during dashboard construction, and missing states were not converted to zero.

4.3 Final ML Targets

The final scope includes 10 hourly KTMB targets—four ETS, two Intercity, one Komuter and three Komuter Utara—and five daily Rapid Rail targets. BRT is excluded from the frozen ML scope.

Service	Station	Tourism destination	Forecast grain
Komuter	Batu Caves	Batu Caves	Hourly
Intercity	Jerantut	Taman Negara	Hourly
Komuter Utara	Padang Besar	Padang Besar / Perlis northern tourism gateway	Hourly
ETS	Seremban	Seremban heritage and Lake Garden	Hourly
ETS	Kuala Kangsar	Royal Town Kuala Kangsar	Hourly
Komuter Utara	Ipoh	Ipoh heritage district / Railway Station	Hourly

Service	Station	Tourism destination	Forecast grain
ETS	Sungai Petani	Bujang Valley Archaeological Heritage	Hourly
Intercity	Dabong	Gunung Stong / Jelawang Waterfall & Dabong caves	Hourly
Komuter Utara	Butterworth	George Town / Penang UNESCO heritage area	Hourly
ETS	Taiping	Taiping heritage / Lake Gardens	Hourly
Rapid Rail	KJ10: KLCC	Petronas Twin Towers / KLCC	Daily
Rapid Rail	KG16: Pasar Seni	Central Market / Petaling Street heritage area	Daily
Rapid Rail	KG15: Muzium Negara	National Museum	Daily
Rapid Rail	KG18: Bukit Bintang	Bukit Bintang / city shopping & entertainment	Daily
Rapid Rail	KJ13: Masjid Jamek	Dataran Merdeka / Sultan Abdul Samad Building	Daily

5. Data Preparation and Feature Engineering

5.1 Cleaning and Coverage

Source schemas were standardised, target identifiers were made explicit, timestamps were parsed at their native grain, and coverage was audited against a complete time grid. Active forecast hours were retained as text metadata. Rapid Rail forecast hours remain blank because its forecasts are daily; no synthetic hour was added.

The model handoff preserves 15 target records, 15 coverage records, 50,981 dashboard rows, 1,162 forecast rows and 105 recommendation rows. Power Query types were aligned with the source meaning: confidence, active forecast hours, dashboard hour and forecast hour are text fields.

5.2 Frozen Feature Set

The implemented feature set includes calendar cycles, target history and lagged demand. Examples in the final feature-importance file include day-of-week sine/cosine terms, week and month cycles, weekend indicator, horizon day, hour cycles for hourly targets, lags at 7, 14, 21 and 28 days, and origin-level recent means, standard deviation, observation count and last value.

No external-event features. Weather, rainfall, temperature, event calendars, public holidays and school holidays are not present in the frozen model. They must not be described as implemented predictors.

5.3 Leakage Control

Lagged and rolling features were constructed from information available before each forecast origin. Evaluation followed chronological partitions, and the test set was not used to select the model. This design reduces temporal leakage and keeps the retrospective test interpretable.

6. Forecasting and Evaluation Methodology

6.1 Candidate Models

Four candidates were compared for every eligible target and horizon: a naive weekly baseline, Linear Regression, Random Forest and XGBoost. The baseline predicts from the corresponding weekly history and remains a first-class candidate, not merely a reporting comparator.

6.2 Direct Multi-Horizon Design

Separate direct forecasts were generated for Day 1 through Day 7. Direct modelling avoids recursively

feeding earlier predictions into later horizons. The chronological workflow was Train → Validation → Test. Validation WAPE selected the winner, after which the locked selection was assessed on the retrospective test period.

6.3 Evaluation Metrics

Mean Absolute Error (MAE) reports the average absolute miss in passenger movements. Root Mean Squared Error (RMSE) places more weight on large misses. Weighted Absolute Percentage Error (WAPE) enables relative comparison across targets with different volumes:

$$\text{WAPE} = \Sigma |y_t - \hat{y}_t| / \Sigma |y_t|$$

WAPE is displayed as a percentage in the report and dashboard; for example, 0.04 is 4%. Improvement over baseline is positive only when the selected model has lower test WAPE than the weekly baseline.

6.4 Demand Classes

Class	Historical percentile rule	Interpretation
Normal	Forecast < Q75	Within the typical historical range
Elevated	$Q75 \leq \text{forecast} < Q90$	Higher than typical demand
Unusually High	Forecast $\geq Q90$	Upper-tail historical demand

Thresholds are target-specific historical percentiles. They are descriptive categories, not probabilities and not universal capacity limits.

6.5 Recommendation Logic

For hourly KTMB targets, the recommendation identifies the lowest predicted rail-demand hour within the eligible forecast window and reports both date and time. For daily Rapid Rail targets, it identifies the lowest predicted day and displays no invented hour. Every recommendation includes the reminder to check attraction opening hours and travel connections before planning a visit.

7. Results and Findings

7.1 Final Test Performance

Target	Selected model	Test WAPE	Baseline WAPE	Improvement	Beat baseline?
KG15: Muzium Negara	Naive Weekly	3.00%	3.00%	+0.00%	No
KG16: Pasar Seni	Random Forest	3.97%	5.67%	+29.95%	Yes
KG18: Bukit Bintang	Linear Regression	6.47%	3.85%	-68.19%	No
KJ10: KLCC	Naive Weekly	4.98%	4.98%	+0.00%	No
KJ13: Masjid Jamek	Naive Weekly	4.60%	4.60%	+0.00%	No
Kuala Kangsar	XGBoost	275.90 %	56.39%	-389.25%	No
Seremban	Random Forest	23.81%	26.76%	+11.03%	Yes
Sungai Petani	XGBoost	73.88%	37.94%	-94.73%	No
Taiping	XGBoost	75.75%	37.84%	-100.19%	No
Dabong	XGBoost	227.73 %	77.12%	-195.30%	No
Jerantut	XGBoost	270.89 %	66.17%	-309.40%	No
Batu Caves	XGBoost	114.16 %	28.21%	-304.74%	No
Butterworth	XGBoost	37.40%	22.33%	-67.47%	No
Ipoh	XGBoost	58.27%	29.58%	-97.02%	No
Padang Besar	XGBoost	35.20%	30.88%	-13.99%	No

Model selection produced nine XGBoost targets, three naive-weekly targets, two Random Forest targets and one Linear Regression target. Selection diversity reflects the validation result for each

target; it does not imply that more complex models were generally superior.

7.2 Verified Positive Results

Pasar Seni selected Random Forest and achieved 3.9749% test WAPE versus 5.6742% for the weekly baseline, a 29.95% improvement. Seremban also selected Random Forest and achieved 23.8064% versus 26.7575%, an 11.03% improvement. These were the only two targets that beat the baseline on the retrospective final test.

7.3 Negative and Neutral Results

Three Rapid Rail targets selected the weekly baseline itself, producing zero improvement by definition. Several KTMB targets selected XGBoost on validation but performed materially worse than the weekly baseline on the test period. For example, Batu Caves recorded 114.16% test WAPE versus 28.21% for the baseline. These results demonstrate instability and support the dashboard's target-specific performance warning.

7.4 Horizon Stability

Performance varies across Day 1–Day 7 horizons. Later horizons are not guaranteed to deteriorate monotonically, and some target–horizon combinations show sharp WAPE increases. The dashboard therefore retains the horizon selector and comparison chart rather than summarising accuracy with one universal claim.

7.5 Data Recency

KTMB source data ends on 8 September 2026 and the final prototype forecasts cover 9–15 September 2026. Rapid Rail source data ends on 24 August 2026 and its forecasts cover 25–31 August 2026. Because the Rapid Rail outputs are older than the final September reporting context, they are labelled stale prototypes and must not be presented as current September demand.

8. Power BI Decision-Support Dashboard

The completed Power BI project contains five pages. Shared navigation and consistent selectors connect the tourism, forecast, recommendation and evaluation layers, while page-specific annotations communicate grain and recency.

8.1 Executive Overview

Combines the Top State, destination forecast, lower-demand recommendation and model-performance snapshot. The tourism metric selector controls only the tourism KPI, map and ranking; the destination selector drives the mobility visuals.

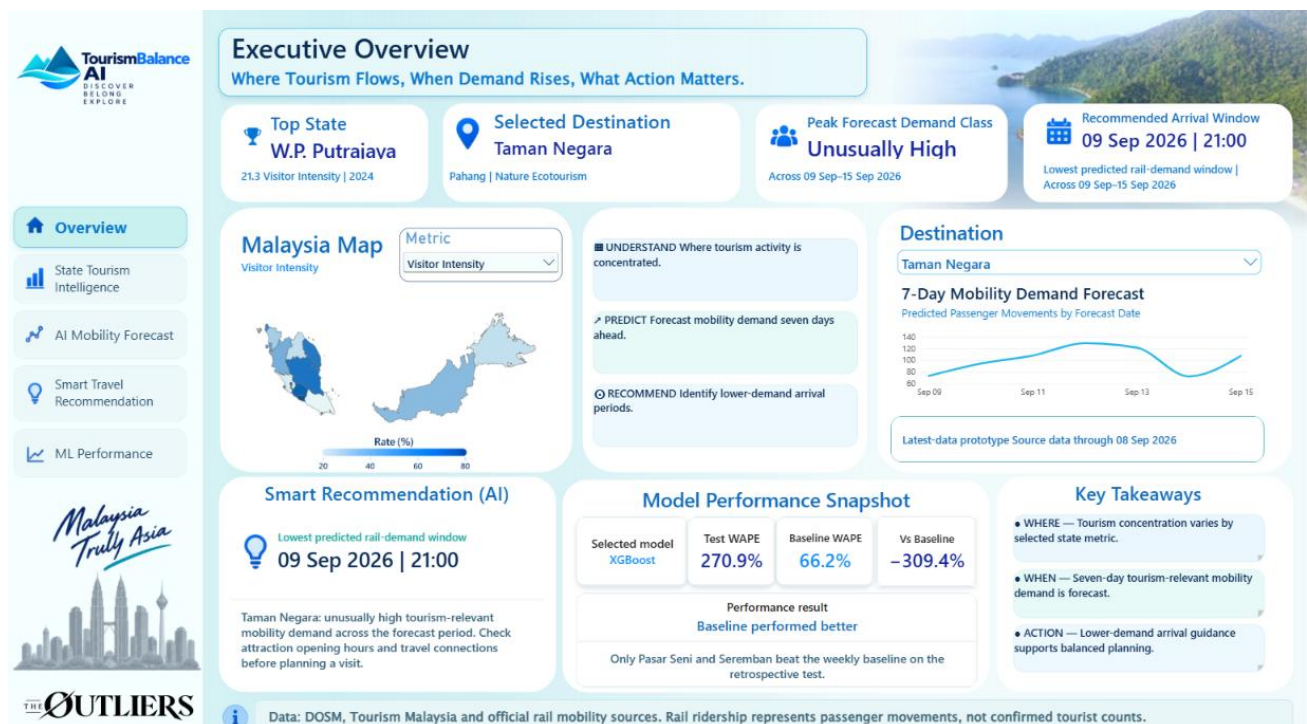


Figure 1. Executive Overview (implemented Power BI page).

8.2 State Tourism Intelligence

Shows Domestic Visitors, Hotel Occupancy Rate and Visitor Intensity through a metric-aware KPI, Malaysia shape map, state ranking and historical trend. Tourism Expenditure is not part of the final implemented selector.

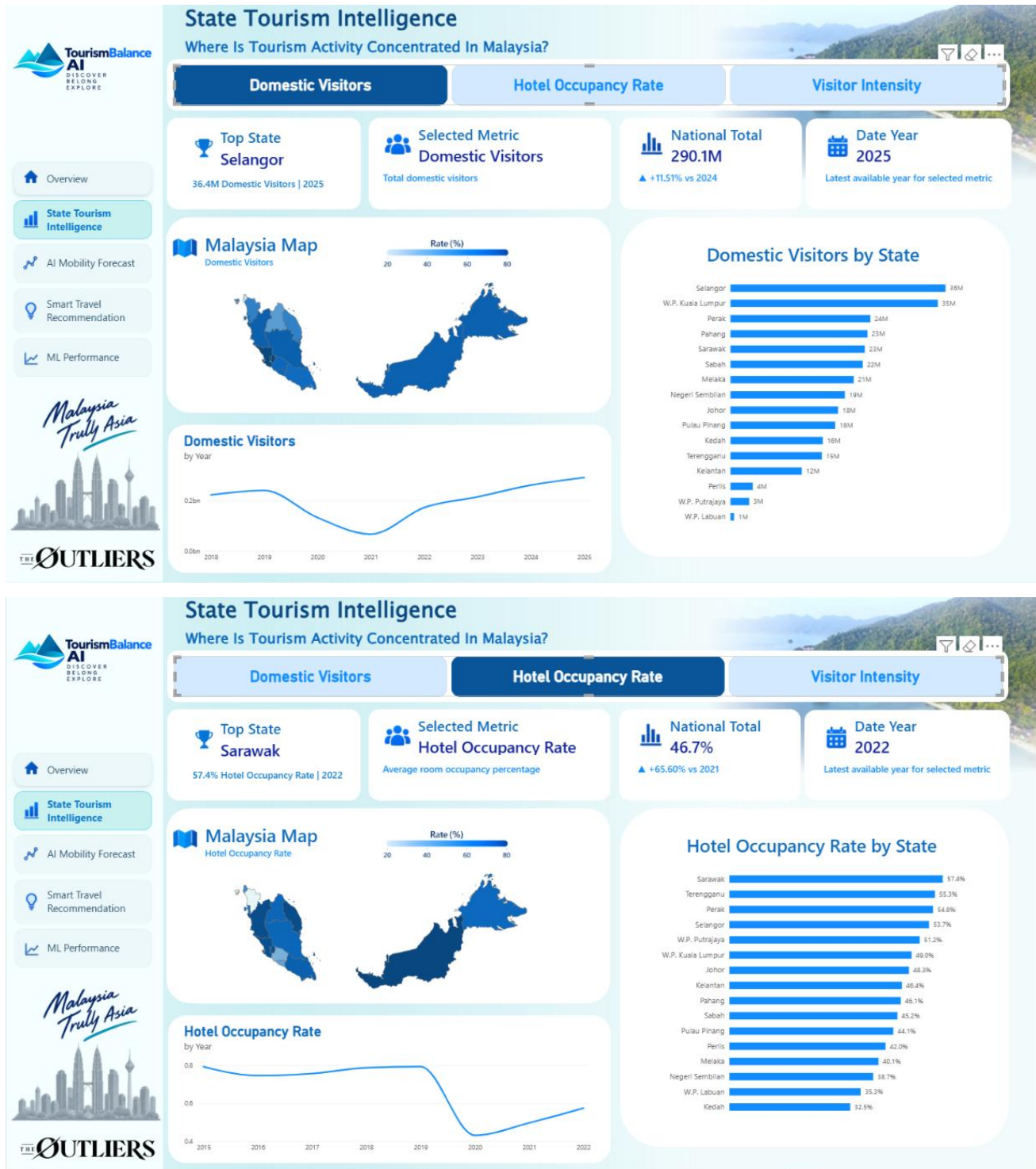




Figure 2. State Tourism Intelligence (implemented Power BI page).

8.3 AI Mobility Demand Forecast

Supports destination, service, forecast-date and horizon exploration. KTMB targets expose hourly forecasts; Rapid Rail remains daily. Source cutoff and recency status are visible.

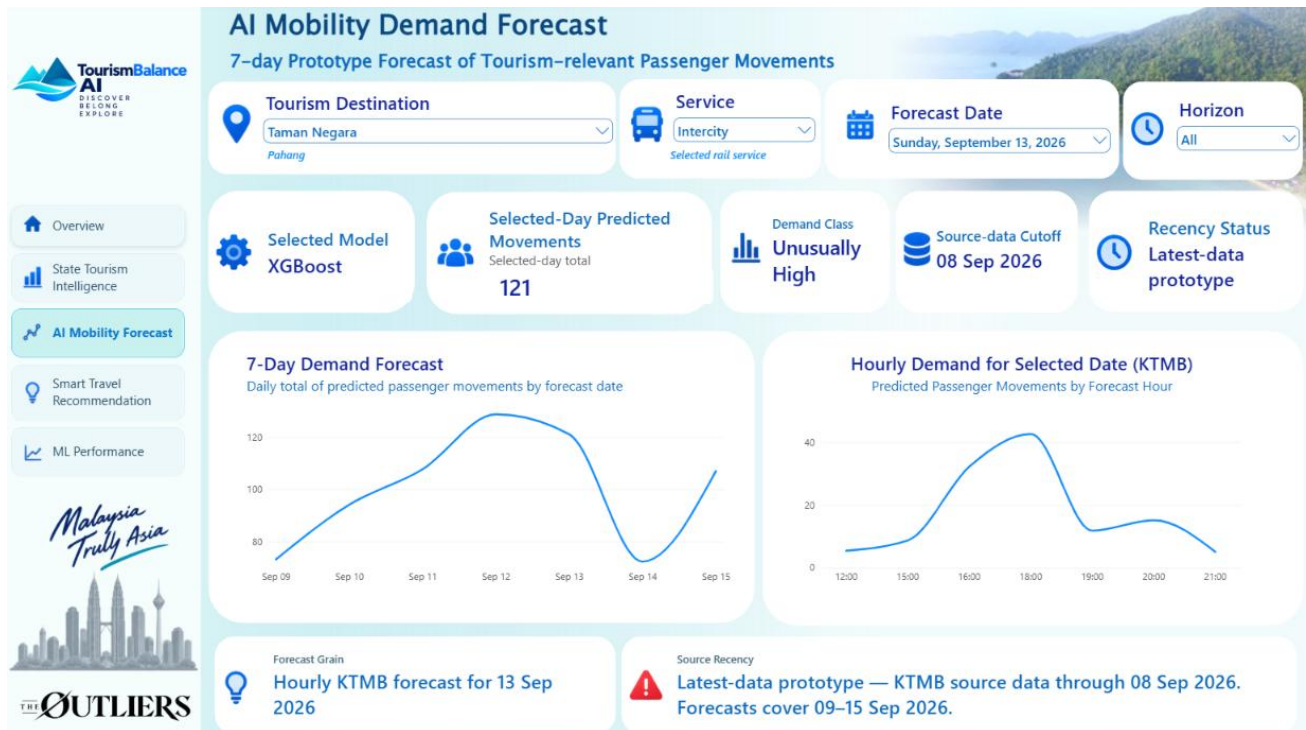


Figure 3. AI Mobility Demand Forecast (implemented Power BI page).

8.4 Smart Travel Recommendation

Displays the day maximum class, the recommended-period class, high-demand periods and the lower-demand arrival window or day, with a practical planning reminder.

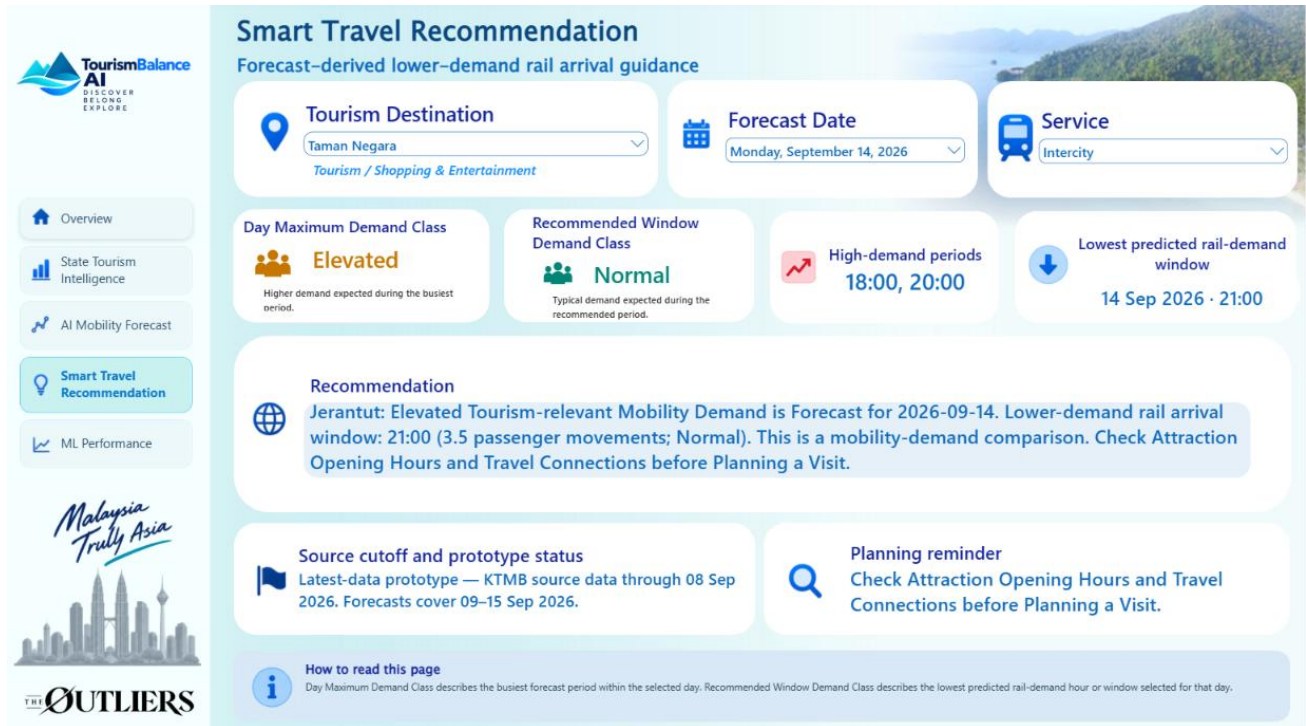


Figure 4. Smart Travel Recommendation (implemented Power BI page).

8.5 ML Performance & Model Evaluation

Reports the selected model, test and baseline WAPE, improvement status, actual-versus-predicted history, horizon comparison and pooled feature-importance availability.

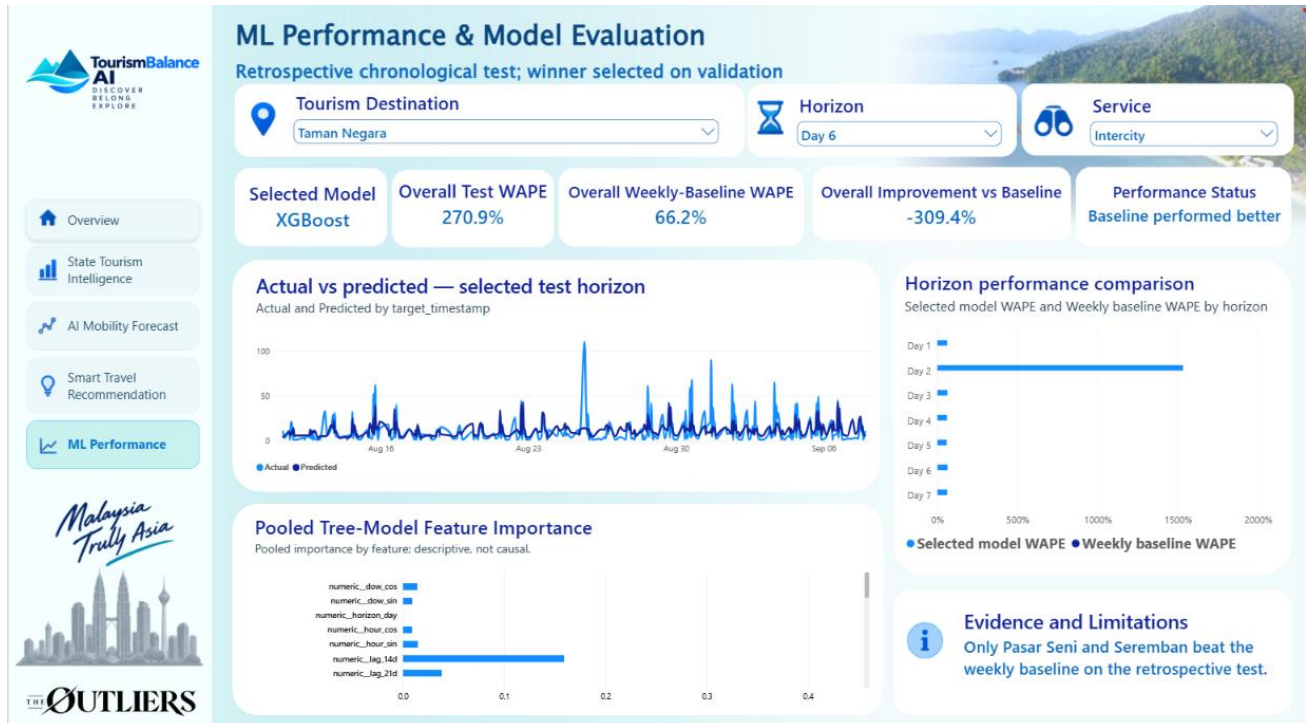


Figure 5. ML Performance & Model Evaluation (implemented Power BI page).

9. Feature Importance

Feature importance is available only for tree-based models and is summarised across pooled combinations of frequency, horizon and model. It is not station-specific and must not be interpreted causally. Linear Regression coefficients and the naive weekly baseline are not directly comparable to tree gain or split importance.

The retained features show that recent origin history, multi-week lags and calendar cycles are used by the forecasting pipeline. The dashboard presents feature importance as optional diagnostic evidence, not as proof that a feature causes ridership demand.

Scope label. *Pooled frequency/horizon/model importance; not station-specific; not causal.*

10. Limitations

Passenger movements are not confirmed tourist counts; destination links establish relevance, not passenger purpose.

Only 15 mapped targets enter the frozen ML evaluation, so results should not be generalised to the entire national rail network. The modelling history consists primarily of 2026 year-to-date observations, which limits the models' ability to learn longer-term trends and recurring annual seasonal patterns.

Rapid Rail is daily while KTMB targets are hourly, limiting direct grain-to-grain comparison.

Rapid Rail forecasts are stale prototypes based on data through 24 August 2026.

Only Pasar Seni and Seremban beat the weekly baseline on the retrospective final test.

Several hourly target-horizon combinations, particularly some Day-2 XGBoost forecasts, showed substantial deterioration in retrospective evaluation. The available evidence does not establish a single causal explanation for this instability, and validation selection did not guarantee test superiority.

Mapping confidence is evidence-based but still involves judgement, especially for gateway and proximity relationships.

Weather, rainfall, temperature, events, public holidays and school holidays are absent from the frozen model.

Feature importance is pooled and associational, not causal or station-specific.

11. Future Work

Automate authoritative data refreshes and expose freshness checks before forecast publication.

Collect passenger-purpose or visitor-survey evidence to calibrate the tourism share of ridership.

Expand mapped targets only after repeatable evidence and confidence review.

Evaluate weather, event, public-holiday and school-holiday features once verified, timely sources are available.

Investigate target-specific fallback rules that retain the weekly baseline when validation gains do not remain stable.

Add uncertainty intervals and monitoring for data drift, coverage failures and horizon degradation.

Assess recommendation usefulness with tourism operators, transport planners and visitors.

12. Conclusion

TourismBalance AI delivers an honest end-to-end prototype for dynamic tourism resilience and demand intelligence. It connects state-level tourism concentration with mapped rail-demand forecasts, converts those forecasts into lower-demand arrival guidance, and exposes model evidence alongside every decision layer.

The study's most important result is not that machine learning always wins. Only Pasar Seni and Seremban outperformed the weekly baseline on the retrospective test, while several targets showed substantial degradation. Preserving those results—together with native data grain, source cutoffs and the passenger-movement interpretation—makes the system more credible and more useful as a foundation for future development.

References

- Department of Statistics Malaysia. (2025). Domestic Tourism Survey 2024. https://www.dosm.gov.my/site/downloadrelease?admin_view=&id=domestic-tourism-survey-2024&lang=English
- Department of Statistics Malaysia. (2025). Domestic Tourism Survey (States) 2024. <https://www.dosm.gov.my/site/downloadrelease?id=domestic-tourism-survey-states-2024&lang=English>
- data.gov.my. (2026). KTMB data catalogue. <https://data.gov.my/data-catalogue?source=KTMB>
- data.gov.my. (2026). KTMB Explorer. <https://data.gov.my/dashboard/ktmb-explorer>
- data.gov.my. (2026). Daily public transport ridership. https://data.gov.my/data-catalogue/ridership_headline
- MyRapid. (2026). Rail routes. <https://myrapid.com.my/rail/routes>
- Tourism Malaysia. (n.d.). Batu Caves. <https://www.malaysia.travel/index.php/explore/batu-caves>
- Tourism Malaysia. (n.d.). Taman Negara. <https://www.malaysia.travel/explore/taman-negara>
- Tourism Malaysia. (n.d.). Petronas Twin Towers. <https://www.malaysia.travel/index.php/explore/petronas-twin-tower>
- Tourism Malaysia. (n.d.). Kuala Lumpur travel guide. <https://ebrochures.malaysia.travel/en/malaysia-travel-guide/kuala-lumpur/>
- Tourism Malaysia. (n.d.). Archaeological heritage tour. <https://www.malaysia.travel/deals/2d1n-archaeological-heritage-tour>
- Tourism Malaysia. (n.d.). Discovering Seremban. <https://www.malaysia.travel/explore/discovering-seremban-through-its-rich-heritage-local-attractions>